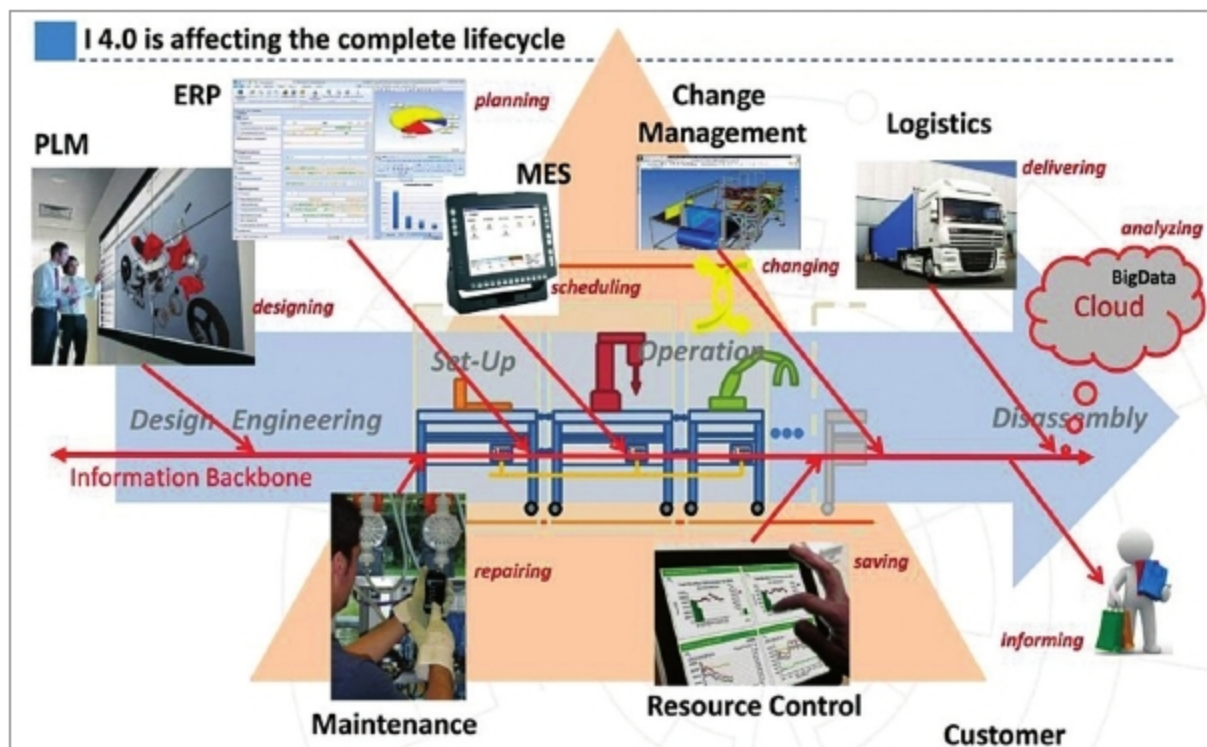




Industry 4.0: The German vision for advanced manufacturing (Credit: <https://image.slidesharecdn.com>)

strategy for the future. It is not only machines and products that are networked in the smart factory; the factory floor of the future is also digital.

Cameras and sensors monitor the automated manufacturing process, and robots intervene and rectify faults in an emergency. Drones fly through the factory and replenish supplies of material. If the warehouse is empty, the software automatically orders more. Workers are nowhere to be seen. This is what the smart factory of the future looks like, and it has already become a reality in some German companies today. Machines, software, and robots are linked together through superfast Internet connections. They communicate, learn from one another, and identify products with the aid of RFID chips.

However, Industry 4.0 is not only about machines manufacturing products autonomously and automatically. The fourth industrial revolution also says farewell to industrial mass production for anonymous customers. Instead, individualised products can be made to order in accordance with the customer's needs. When an order is received at the factory, the software programs the production line. Thanks to artificial intelligence, the systems learn something new every time they process an order.

Manufacturing for specific customers rather than the warehouse is cheaper and more ecological. Since most products are becoming increasingly digitalised, it is also becoming

increasingly rare for a business relationship to end with the sale of a finished product. Instead, hybrid products are being sold, including services such as software updates and maintenance.

The latest German technology for the smart factory is the digital floor. Equipped with electricity and data cables, it can supply and control the robots. That does away with the need for powerful batteries and complex positioning systems. Manufacturing systems can also be controlled through the floor and production lines modified as required. All the components in the smart factory are connected, flexible, and mobile. Only the factory building itself is immovable—for the time being at least.

The smart factory, fast Internet links, and e-health: Some European countries are international leaders.

Luxembourg: Fast Internet

Fast Internet connections are the basic prerequisite for digitalisation. With coverage of practically 100 per cent, Luxembourg has by far the best-developed broadband network in Europe.

The small Grand Duchy lands in second place—behind Singapore—in the Digital Readiness Index published by network specialists, Cisco. The ranking of 141 states shows how well countries are prepared for a digital future.

Luxembourg comes first even in the ranking of the best framework conditions for startups. The govern-

ment of Luxembourg supports new companies and has a dynamic venture capital market.

Luxembourg has invested a large amount of money in its digital infrastructure to ensure that access to broadband Internet is universally available. Coverage with especially fast Internet offering download speeds of over thirty megabits per second is 96 per cent. This makes Luxembourg number one in Europe; the European average is 74 per cent.

Luxembourg achieves an above-average score when it comes to mobile Internet access. There are Wi-Fi hot-spots everywhere in public spaces. This means citizens can go online at all times without using up their mobile data contingent.

Finland: E-Health

A virtual hospital and electronic patient records: Finland's e-health system is one of the most advanced in the world.

Finland has long led Europe when it comes to digital technologies. Following mobile phones, online banking, and electronic government (e-government), digital health has now become the largest high-tech export sector. Prescriptions for prescription-only medicines have been issued electronically since 2010, and electronic patient records have now been used in the Nordic country for over twenty years. They store health data, medical diagnoses, and prescriptions. As a result, patients and doctors who treat them can view blood values or previous illnesses from any location.

If you come from a sparsely populated part of the country and have a health problem, you can visit a virtual health village. The platform offers serious information online and guides users through the Finnish healthcare system. Patients can talk to doctors online and arrange appointments.

The latest sensation in the field of digital health is FinnGen, the national genetic research project launched in 2017. Genetic data is being collected from ten per cent of all Finns. The goal is to make changes in genetic material visible and prevent diseases. **EFY**

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